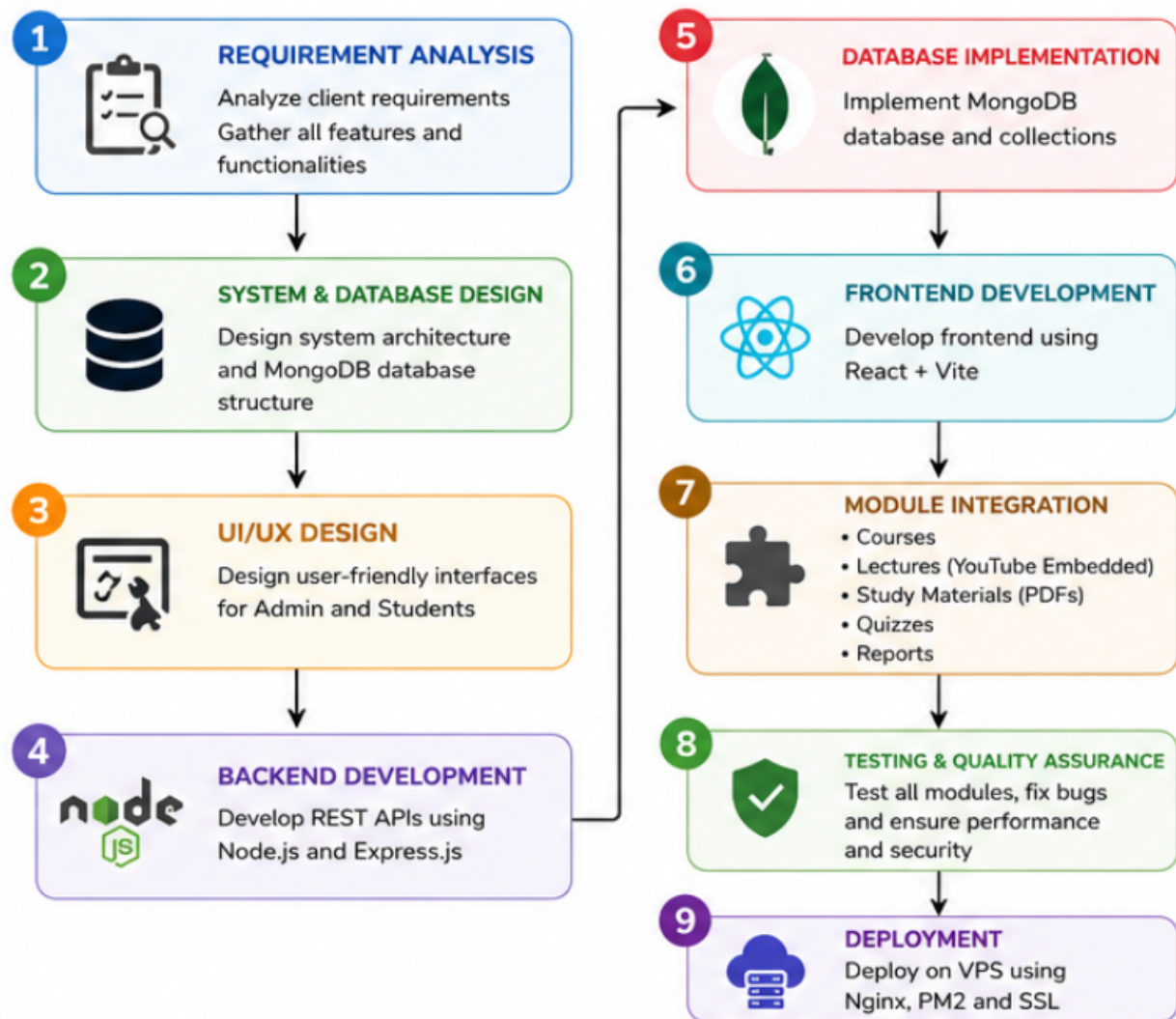


METHODOLOGY DIAGRAM



Phase 1: Requirement Analysis

First, we gather and analyze the client's requirements to understand the learning platform's objectives, target users, and required functionalities.

Activities:

- Understand institute requirements
- Identify admin and student roles
- Define course structure
- Define quiz and learning material requirements
- Estimate expected student capacity

Output:

- Software Requirement Specification (SRS)

Phase 2: System Design

The system architecture, database structure, and workflow are designed to ensure scalability, security, and ease of use.

Activities:

- Design LMS architecture
- Design MongoDB database
- Define user roles and permissions
- Create module structure

Output:

- System Architecture Diagram
- Database Design

Phase 3: UI/UX Design

User-friendly interfaces are designed for both administrators and students.

Activities:

- Admin Dashboard Design
- Student Dashboard Design
- Course Page Design
- Quiz Interface Design
- Responsive Mobile Layout

Output:

- Wireframes and User Interface Design

Phase 4: Development

The LMS platform is developed using modern web technologies.

Frontend Technologies

- React.js
- Vite
- Tailwind CSS

Backend Technologies

- Node.js
- Express.js

Database

- MongoDB

Modules Developed

- Authentication System
- Course Management
- Lecture Management
- Study Materials
- Quiz System

Output:

- Functional LMS Platform

Phase 5: Testing

The system is thoroughly tested to ensure reliability and performance.

Activities:

- Functional Testing
- User Authentication Testing
- Quiz Testing
- Performance Testing
- Security Testing

Output:

- Bug-Free System

Phase 6: Deployment

The completed LMS is deployed on the VPS server and made accessible online.

Server Environment

- Ubuntu VPS
- Nginx
- PM2
- MongoDB
- SSL Certificate

Output:

- Live LMS Website

Phase 7: Training & Support

The administrator is trained to manage courses, lectures, and quizzes effectively.

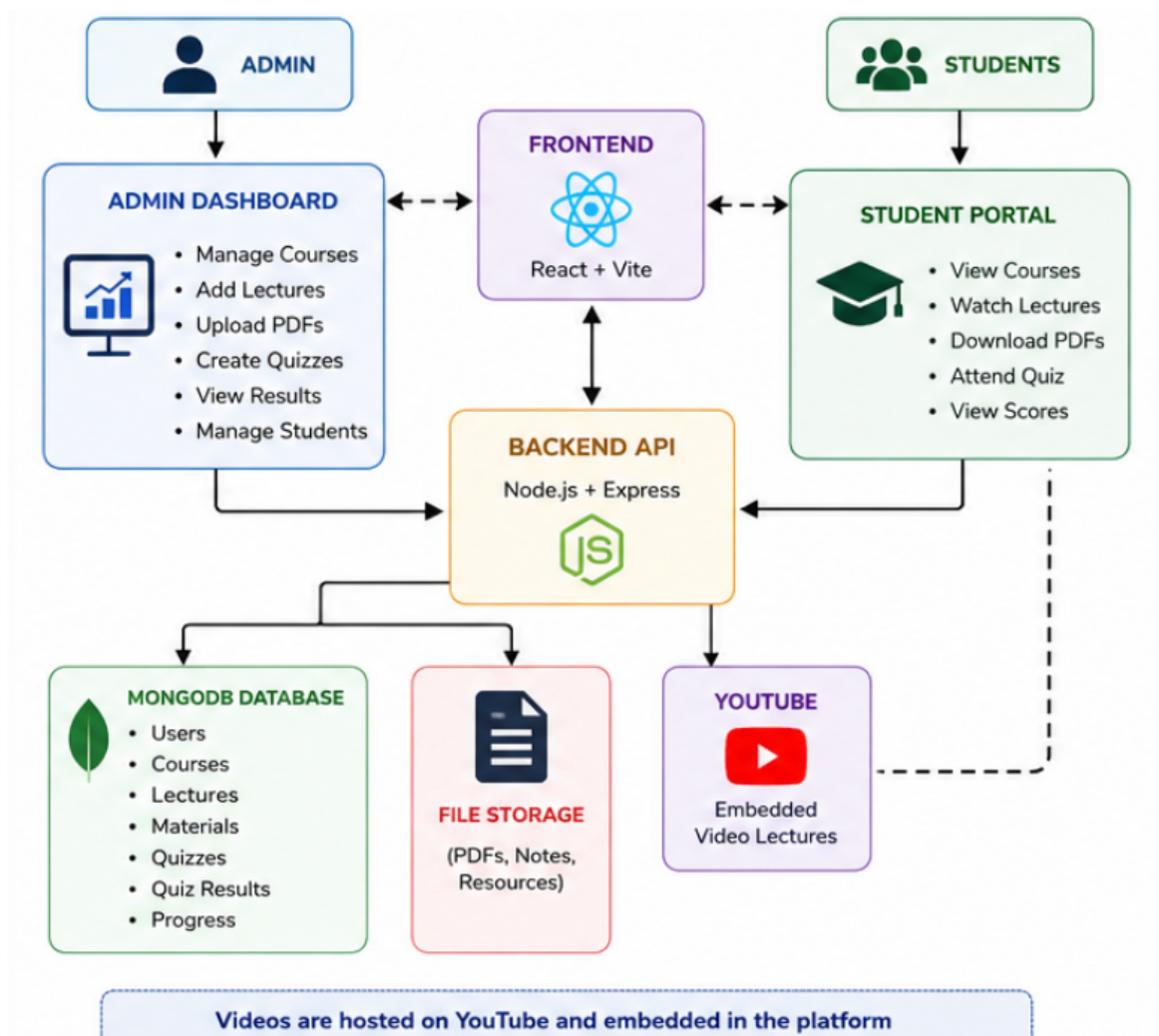
Activities:

- Admin Training
- Documentation
- Technical Support & Maintenance

Output:

- Fully Operational LMS Platform

SYSTEM ARCHITECTURE



Presentation Layer (Frontend)

This is the interface used by Admins and Students.

Technology:

- React.js
- Vite
- Tailwind CSS

Functions:

- Login & Registration
- Course Dashboard
- Watch Lectures
- Attend Quizzes
- Download Study Materials

Application Layer (Backend)

This is the core business logic layer.

Technology:

- Node.js
- Express.js

Functions:

- User Authentication
- Course Management
- Quiz Evaluation
- Student Management
- API Handling
- Access Control

Data Layer (Database)

All LMS data is stored here.

Technology:

- MongoDB

Stores:

- Users
- Courses
- Lectures
- PDFs
- Quizzes

TECHNOLOGIES AND TOOLS USED

Frontend Technologies

Technology	Purpose
React.js	User Interface Development
Vite	Fast Frontend Build Tool
Tailwind CSS	Responsive UI Design
Axios	API Communication
React Router	Navigation and Routing

Backend Technologies

Technology	Purpose
Node.js	Server-Side Runtime Environment
Express.js	REST API Development
JWT	Authentication & Authorization
bcrypt.js	Password Encryption
Multer	File Upload Management

Database

Technology	Purpose
MongoDB	NoSQL Database
Mongoose	MongoDB Object Modeling

Security Technologies

Technology	Purpose
JWT Authentication	Secure Login
bcrypt.js	Password Hashing
Role-Based Access Control	Admin & Student Permissions
HTTPS/SSL	Secure Data Transmission